SEPARATEDNESS APPEARS AT A FINITE STAGE IN THE DESCENT SITUATION

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Editorial note. Counted span is the author's own proof environment, source lines 1553-1568, following the statement at 1540-1551.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Etale neighbourhood covering a quasi-compact image. constructed within the proof (~1553-1558). Affine W -> Y etale whose image contains the image of |X_0| -> |Y|; the diagonal of W x_Y X_i -> W is the base change of the diagonal of X_i -> X_i x_Y X_i by the surjective etale (X_i x_Y X_i) x_Y W -> X_i x_Y X_i.
- Scheme-level descent of separatedness. lemma-descend-separated (same chapter, cited at ~1567-1568). Affine case engine: X separated gives X_i separated for i large.

Source reference. Original lemma 084T: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1336–1336. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let Y be an algebraic space over S. Let $X = \lim X_i$ be a directed limit of algebraic spaces over Y with affine transition morphisms. Assume*

(1) *Y is quasi-separated,*
(2) *X_i is quasi-compact and quasi-separated,*
(3) *the morphism $X \to Y$ is separated.*

Then $X_i \to Y$ is separated for all i large enough.

Proof. Let $0 \in I$. Choose an affine scheme W and an étale morphism $W \to Y$ such that the image of $|W| \to |Y|$ contains the image of $|X_0| \to |Y|$. This is possible as X_0 is quasi-compact. It suffices to check that $W \times_Y X_i \to W$ is separated for some $i \geq 0$ because the diagonal of $W \times_Y X_i$ over W is the base change of $X_i \to X_i \times_Y X_i$ by the surjective étale morphism $(X_i \times_Y X_i) \times_Y W \to X_i \times_Y X_i$. Since Y is quasi-separated the algebraic spaces $W \times_Y X_i$ are quasi-compact (as well as quasi-separated). Thus we may base change to W and assume Y is an affine scheme. When Y is an affine scheme, we have to show that X_i is a separated algebraic space for i large enough and we are given that X is a separated algebraic space. Thus this case follows from Lemma 084T. □

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.